

Rajalakshmi Engineering College

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Batch: 2028

Degree: B.E - CSD

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2024_28_IV_CSD_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 7_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Maria, an online store owner, is looking to implement a pricing system that calculates the final price of products after applying discounts. She needs a program that takes the original price of a product and the discount percentage as input and computes the final discounted price. The discount is applied as a percentage of the original price. Maria wants to ensure that the final price is formatted to display exactly two decimal places.

Implement this functionality using the PriceCalculator interface and the DiscountCalculator class.

Input Format

The first line of input consists of the original price (a double value).

The second line of input consists of a discount percentage (a double value).

Output Format

The output displays the final price after the discount, adhering to the following format: "Final Price after discount: \$[final_price]".

Here, [final_price] should be replaced with the calculated final price, formatted as a currency value with two decimal places.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 100.0
10.0

Output: Final Price after discount: \$90.00

Answer

```
import java.util.Scanner;

interface PriceCalculator {
    double calculatePrice(double originalPrice, double discountPercentage);
}

class DiscountCalculator implements PriceCalculator {
    public double calculatePrice(double originalPrice, double discountPercentage)
    {
        return originalPrice - (originalPrice * discountPercentage / 100);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        double originalPrice = scanner.nextDouble();
        double discount = scanner.nextDouble();
        PriceCalculator calculator = new DiscountCalculator();
        double finalPrice = calculator.calculatePrice(originalPrice, discount);
        System.out.printf("Final Price after discount: $%.2f\n", finalPrice); //
        // Formats output to 2 decimal places
    }
}
```

}

Status : Correct

Marks : 10/10

2. Problem Statement

John is developing a car loan calculator and has structured his program using two interfaces, Principal and InterestRate, defining methods for principal and interest rate retrieval.

The Loan class implements these interfaces, taking principal and annual interest rates as parameters. User input is solicited for these values, and the program ensures their validity before performing calculations. If input values are invalid (less than or equal to zero), an error message is displayed.

Note: Total interest = principal * interest rate * years

Input Format

The first line of input consists of a double value P, representing the principal.

The second line consists of a double value R, representing the annual interest rate.

The third line consists of an integer value N, representing the loan duration in years.

Output Format

If the input values are valid, print "Total interest paid: Rs. " followed by a double value, representing the total interest paid, rounded off to two decimal places.

If the input values are invalid (negative or zero values for principal, annual interest rate, or loan duration), print "Invalid input values!".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 20000.00

0.05

5

Output: Total interest paid: Rs.5000.00

Answer

```
import java.util.Scanner;
```

```
interface Principal {  
    double getPrincipal();  
}
```

```
interface InterestRate {  
    double getInterestRate();  
}
```

```
class Loan implements Principal, InterestRate {  
    double principal;  
    double annualRate;
```

```
    Loan(double principal, double annualRate) {  
        this.principal = principal;  
        this.annualRate = annualRate;  
    }
```

```
    public double getPrincipal() {  
        return principal;  
    }
```

```
    public double getInterestRate() {  
        return annualRate;  
    }
```

```
    public double calculateTotalInterest(int years) {  
        return principal * annualRate * years;  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);
```

```
        double carPrice = scanner.nextDouble();
```

```

double annualInterestRate = scanner.nextDouble();

int loanDuration = scanner.nextInt();

if (carPrice <= 0 || annualInterestRate <= 0 || loanDuration <= 0) {
    System.out.println("Invalid input values!");
    return;
}

Loan carLoan = new Loan(carPrice, annualInterestRate);
double totalInterest = carLoan.calculateTotalInterest(loanDuration);

System.out.printf("Total interest paid: Rs.%.2f%n", totalInterest);
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement:

Sam is developing a geometry application and needs a class for trapezoid calculations. Create a "Trapezoid" class implementing a "ShapeInput" interface with a method to input trapezoid dimensions.

Also, implement a "ShapeCalculator" interface with methods to compute area and perimeter. In the "Main" class, instantiate Trapezoid, gather user input, and display the calculated area and perimeter with two decimal places.

Note

Area of Trapezoid = $(1/2) * (base1 + base2) * height$

Perimeter of Trapezoid = base1 + base2 + side1 + side2

Input Format

The first line of input is a double-point value representing base1 of the trapezoid.

The second line of input is a double-point value representing base2 of the trapezoid.

The third line of input is a double-point value representing the height of the trapezoid.

The fourth line of input is a double-point value representing side1 of the trapezoid.

The fifth line of input is a double-point value representing side2 of the trapezoid.

Output Format

The output displays the two lines of the calculated area (double type) and perimeter (double type) of the trapezoid, each rounded to two decimal places in the following format:

"Area of the Trapezoid: <<calculated area>>".

Perimeter of the Trapezoid: <<calculated perimeter>>".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 1.0

2.0

1.0

3.0

1.0

Output: Area of the Trapezoid: 1.50

Perimeter of the Trapezoid: 7.00

Answer

```
import java.util.Scanner;
```

```
interface ShapeInput {  
    void getInput();
```

```
}
```

```
interface ShapeCalculator {  
    double calculateArea();  
    double calculatePerimeter();  
}
```

```
class Trapezoid implements ShapeInput, ShapeCalculator {  
    double base1, base2, height, side1, side2;
```

```
    public void getInput() {  
        Scanner sc = new Scanner(System.in);  
        base1 = sc.nextDouble();  
        base2 = sc.nextDouble();  
        height = sc.nextDouble();  
        side1 = sc.nextDouble();  
        side2 = sc.nextDouble();  
    }
```

```
    public double calculateArea() {  
        return 0.5 * (base1 + base2) * height;  
    }
```

```
    public double calculatePerimeter() {  
        return base1 + base2 + side1 + side2;  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Trapezoid trapezoid = new Trapezoid();  
        trapezoid.getInput();
```

```
        double area = trapezoid.calculateArea();  
        double perimeter = trapezoid.calculatePerimeter();
```

```
        System.out.println("Area of the Trapezoid: " + String.format("%.2f", area));  
        System.out.println("Perimeter of the Trapezoid: " + String.format("%.2f",  
perimeter));  
    }  
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Alex and Bob are designing a control system for household appliances, and one of the appliances is a washing machine. You want to create a program to help them that models the washing machine as a motor and calculates its electricity consumption based on its capacity.

Define an interface named `Motor` with the following methods:

```
void run() double consume(double capacity)
```

Create a class called `WashingMachine` that implements the `Motor` interface.

In the `WashingMachine` class:

Implement the `run()` method to print "Washing machine is running." Implement a `consume()` method to print "Washing machine is consuming electricity." Implement the `consume(double capacity)` method to calculate the electricity consumption (in kWh) of the washing machine based on its capacity. The formula for electricity consumption is $(\text{capacity} * 0.05)$.

Input Format

The input consists of a double value representing the capacity of the washing machine in kW.

Output Format

The first line of output prints "Washing machine is running."

The second line prints "Washing machine is consuming electricity."

The third line prints "Electricity consumption: X kWh" where X is a double value, rounded off to two decimal places, representing the electricity consumption.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2.5

Output: Washing machine is running.
Washing machine is consuming electricity.
Electricity consumption: 0.13 kWh

Answer

```
import java.util.Scanner;

interface Motor {
    void run();
    void consume();
    double consume(double capacity);
}

class WashingMachine implements Motor {
    public void run() {
        System.out.println("Washing machine is running.");
    }

    public void consume() {
        System.out.println("Washing machine is consuming electricity.");
    }

    public double consume(double capacity) {
        return capacity * 0.05;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        WashingMachine washingMachine = new WashingMachine();

        double capacity = scanner.nextDouble();

        washingMachine.run();
        washingMachine.consume();

        double consumption = washingMachine.consume(capacity);
        System.out.printf("Electricity consumption: %.2f kWh", consumption);

        scanner.close();
    }
}
```

}
}

Status : Correct

Marks : 10/10